

Unit 3 Problems

(1)

- 1) A dataset has a binary target and a single numeric feature X . There are 8 samples $X = [1, 2, 3.5, 5, 5.4, 6, 7, 9.8]$ and corresponding labels 00001111 . If we consider a single split of the form $X \leq t$ in a decision tree, which threshold ' t ' will minimize the Gini impurity at the root?

Solution:

X	1	2	3.5	5	5.4	6	7	9.8
Labels	0	0	0	0	1	1	1	1

To minimize the Gini impurity, we want the split to make both sides as pure as possible

If we choose a threshold between 5 and 5.4

$$t = \frac{5 + 5.4}{2} = \frac{10.4}{2} = 5.2$$

Then Left side ($X \leq 5.2$)

1, 2, 3.5, 5 $\Rightarrow$ all 0's Gini = 0

Right side ($X > 5.2$)

~~The remaining 9.~~

5.4, 6, 7, 9.8 $\Rightarrow$ all 1's

Gini = 0

optimal threshold $t = 5.2$

So that Gini impurity = 0

- 2) You are building a decision tree classifier using entropy as the impurity measure. At a node, there are 10 samples: 6 of class A and 4 of class B. What is the entropy of this node (base 2) rounded to three decimal places.

Solution:

$$H = - [p_1 \log_2 p_1 + p_2 \log_2 p_2]$$

$$H = - [0.6 \log_2 0.6 + 0.4 \log_2 0.4]$$

$$H = 0.917 \approx 0.917$$

$p_1 = \frac{6}{10}$
 $p_2 = \frac{4}{10}$

- 3) Consider a node in a classification tree with 12 samples: 3 of class 0, 3 of class 1, and 6 of class 2. What is the Gini impurity of this node

Solution:

$$p_0 = \frac{3}{12} = 0.25 \quad p_1 = \frac{3}{12} = 0.25 \quad p_2 = \frac{6}{12} = 0.5$$

$$\text{Gini Index} = 1 - \sum_i p_i^2$$

$$= 1 - [0.25^2 + 0.25^2 + 0.5^2]$$

$$G = 0.625$$

- 4) In a regression tree node, there are 6 observations with target values [2, 4, 4, 5, 7, 8]. You are considering a split on a numeric feature that divides these into left child [2, 4, 4] and right child [5, 7, 8]. What is the reduction in total sum of squared errors (SSE) achieved by this split compared to no split?

Solution:

$$\text{SSE}_{\text{no split}} = \frac{1}{6} \sum_{i=1}^6 (x_i - \bar{x})^2$$

$$\text{SSE}_{\text{split}} = \frac{1}{3} \sum_{i=1}^3 (x_i - \bar{x}_L)^2 + \frac{1}{3} \sum_{i=4}^6 (x_i - \bar{x}_R)^2$$

$$\text{Reduction} = \text{SSE}_{\text{no split}} - \text{SSE}_{\text{split}}$$

(2a)

4) In a Regression tree node, there are 6 observations with target values $[2, 4, 4, 5, 7, 8]$. You are considering a split on a numeric feature that divides these into left child $[2, 4, 4]$ and right child $[5, 7, 8]$, what is the reduction in total Sum Squared errors (SSE) achieved by this split compared to no split?

Solution: Data = $[2, 4, 4, 5, 7, 8]$

$$\text{Mean} = \frac{30}{6} = 5$$

$$\begin{aligned} \text{SSE} &= (2-5)^2 + (4-5)^2 + (4-5)^2 + (5-5)^2 + (7-5)^2 + (8-5)^2 \\ &= 24 \end{aligned}$$

For left child: data = $[2, 4, 4]$

$$\text{Mean} = \frac{10}{3} = 3.33$$

$$\begin{aligned} \text{SSE} &= (2-3.33)^2 + (4-3.33)^2 + (4-3.33)^2 \\ &= 2.66 \end{aligned}$$

For right child: data = $[5, 7, 8]$

$$\text{Mean} = \frac{20}{3} = 6.67$$

$$\begin{aligned} \text{SSE} &= (5-6.67)^2 + (7-6.67)^2 + (8-6.67)^2 \\ &= 4.67 \end{aligned}$$

Total SSE after split: ~~2.66 + 4.67 = 7.33~~

$$2.66 + 4.67 = 7.33$$

$$\text{Reduction in SSE} : 24 - 7.33 = 16.67$$

[illegible]

Solution:

Given $(x, y) = (1, 0), (2, 0), (3, 1), (8, 1), (9, 1), (10, 0)$

possible thresholds (midpoints)

$$L = \frac{1+2}{2} = 1.5$$

$$L = \frac{2+3}{2} = 2.5$$

$$t = \frac{3+8}{2} = 5.5$$

$$t = \frac{8+9}{2} = 8.5$$

$$t = \frac{9+10}{2} = 9.5$$



Case (i):

$z = 1.5$ Left node $\Rightarrow$ Gini = 0
 Right node [0, 1, 1, 1, 0]

$$p_0 = \frac{2}{5} \quad p_1 = \frac{3}{5}$$

$$p_0 = 0.4 \quad p_1 = 0.6$$

(4)

$$\begin{aligned}
 Gini &= 1 - (p_1^2 + p_2^2) \\
 &= 1 - (0.4^2 + 0.6^2) \\
 &= 1 - (0.16 + 0.36) \\
 &= 0.48
 \end{aligned}$$

$$\begin{aligned}
 \text{Weighted Gini} &= \frac{1}{6} \times 0 + \frac{5}{6} \times 0.48 \\
 &= 0.40
 \end{aligned}$$

Case (ii):

$$t = 2.5$$

$$\text{Left node } [0, 0] \Rightarrow Gini = 0$$

$$\text{Right node } [1, 1, 0]$$

$$p_0 = \frac{1}{4} \quad p_1 = \frac{3}{4}$$

$$\begin{aligned}
 Gini &= 1 - (p_0^2 + p_1^2) \\
 &= 0.375
 \end{aligned}$$

$$\begin{aligned}
 \text{Weighted Gini} &= 0 \times \frac{2}{6} + \frac{4}{6} \times 0.375 \\
 &= 0.25
 \end{aligned}$$

Case (iii):

$$t = 5.5$$

$$\begin{aligned}
 \text{Left node } [0, 0, 1] \quad Gini &= 1 - \left(\frac{2}{3}\right)^2 - \left(\frac{1}{3}\right)^2 \\
 &= 0.4444
 \end{aligned}$$

$$\begin{aligned}
 \text{Right node } [1, 0] \quad Gini &= 0.4444 \\
 &= 0.5
 \end{aligned}$$

$$\begin{aligned}
 \text{Weighted gini} &= \frac{3}{6} \times 0.4444 + \frac{3}{6} \times 0.4444 \\
 &= 0.4444
 \end{aligned}$$

Case (iv):

$$t = 8.5$$

$$\begin{aligned}
 \text{Left node } [0, 0, 1, 1] \Rightarrow Gini &= 1 - \left(\frac{2}{4}\right)^2 - \left(\frac{2}{4}\right)^2 \\
 &= 0.5
 \end{aligned}$$

$$\text{Right node } [1, 0] \Rightarrow Gini = 0.5$$

$$\begin{aligned}
 \text{Weighted gini} &= \frac{4}{6} \times 0.5 + \frac{2}{6} \times 0.5 = 0.5 \\
 &= 0.5 \left[\frac{4+2}{6} \right] = 0.5
 \end{aligned}$$

(5)

Case (v): $t = 9.5$

Left node: $[0, 0, 1, 1, 1]$

$$Gini = 1 - \left(\frac{2}{5}\right)^2 - \left(\frac{3}{5}\right)^2 = 0.48$$

Right node: $[0]$ $Gini = 0$

Weighted Gini = ~~$\frac{5}{6} \times 0.48 + \frac{1}{6} \times 0$~~

$$= \frac{5}{6} \times 0.48 + \frac{1}{6} \times 0 = \frac{5 \times 0.48}{6} = 0.4$$

Best threshold ~~$t = 2.5$~~

Minimum weighted Gini impurity = 0.25

- 6) A classification tree uses misclassification error (0-1 loss) as its split criterion. At a node with 50 samples, 30 belongs to class-0 and 20 belongs to class-1. If the node is turned into a leaf (no split), what is the misclassification error of this leaf, and ^{what} class label will it predict?

Solution:

class 0: 30 samples

class 1: 20 samples

$$p_0 = \frac{30}{50} \quad p_1 = \frac{20}{50}$$

$$\text{Misclassification Error} = 1 - \max(p_i)$$

$$= 1 - \frac{30}{50}$$

$$= \frac{20}{50} = 0.4$$

Predicted class = 0.6

- 7) You use a histogram with equal-width bins of width $h=1$ to estimate a density from 100 data points. one bin (from 3 to 4) contains 20 points. What is the estimated density value in that bin?

Solution:

$$\hat{p}(x) = \frac{\# \{x^i \text{ in the same bin as } x\}}{Nh}$$

$$= \frac{20}{100 \times 1} = \underline{0.2}$$

- 8) In Kernel density estimation with a Gaussian kernel, you have $n=4$ data points: $x = \{0, 1, 2, 3\}$ and bandwidth $h=1$ what is the Kernel contribution at $x=1$ from the data point at 1? (Use the Gaussian kernel $K(u) = \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}}$)

Solution:

$$K(u) = \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}}$$

$$x=1 \quad x_i=1 \quad h=1$$

$$K(u) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(1-1)^2}{2}} = \frac{1}{\sqrt{2\pi}}$$

$$\text{Kernel Contribution} = \frac{1}{\sqrt{2\pi}} = \underline{0.3999}$$

9) A Kernel density estimator is defined as

$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x-x_i}{h}\right)$. What happens numerically to the estimated density if you reduce the bandwidth h while keeping all else fixed?

Solution: $\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x-x_i}{h}\right)$

When h is reduced, makes kernel taller and density becomes ~~sped~~ spiky.

10) You have a 1D dataset with labeled points
class A: $x = \{1, 2, 2.5\}$, class B: $\{4, 5, 6\}$ using
a 3-nearest neighbor classifier with Euclidean
distance, what class will a new point at $x=3$
be assigned to?

Solution:

Sample	class	Distance from sample $x=3$
1	A	2
2	A	1
2.5	A	0.5
4	B	1
5	B	2
6	B	3

Ascending order distance

$x=3$ belongs to Class A

$(0.5, A)$
 $(1, A)$
 $(1, B)$
 $(2, A)$
 $(2, B)$
 $(3, B)$

- 11) Consider two dimensional data with three labeled points: $P_1 = (0, 0)$ labeled class-1, $P_2 = (0, 3)$ labeled class 2, and $P_3 = (4, 0)$ labeled class 2. Using 1-nearest neighbor with Euclidean, what class will a new point at $(1, 1)$ be assigned to?

Solution:

Sample	class	Distance from $(1, 1)$
$(0, 0)$	1	$\sqrt{2} = 1.4142$
$(0, 3)$	2	$\sqrt{1^2 + 2^2} = \sqrt{5} = 2.2360$
$(4, 0)$	2	$\sqrt{3^2 + 1^2} = \sqrt{10} = 3.1622$

Point $(1, 1)$ belongs to class-1

- 12) In a 2D space, suppose you have labeled training points: $(0, 0)$ and $(0, 2)$ labeled class A, $(3, 0)$ and $(3, 2)$ labeled class B. Using 3-NN with Euclidean distance, how will a new point at $(1, 1)$ be classified?

Sample	class	distance from $(1, 1)$
$(0, 0)$	A	$\sqrt{2} = 1.4142$
$(0, 2)$	A	$\sqrt{2} = 1.4142$
$(3, 0)$	B	$\sqrt{5} = 2.2360$
$(3, 2)$	B	$\sqrt{5} = 2.2360$
		Arranging distance in ascending order, it can be classified to class-A

Ripper Algorithm

Ripper (Pos, Neg, k)

RuleSet $\leftarrow$ LearnRuleSet (Pos, Neg)

For k times

RuleSet $\leftarrow$ OptimizeRuleSet (RuleSet, Pos, Neg)

LearnRuleSet (Pos, Neg)

RuleSet $\leftarrow \phi$

DL $\leftarrow$ DescLen (RuleSet, Pos, Neg)

Repeat

Rule $\leftarrow$ LearnRule (Pos, Neg)

Add Rule to RuleSet

DL' $\leftarrow$ DescLen (RuleSet, Pos, Neg)

IF DL' > DL + 64 $\rightarrow$ Ripper stops adding rules when the description length of the rule base is more than 64 bits larger than the best description length so far.

PruneRuleSet (RuleSet, Pos, Neg)

Return RuleSet

IF DL' < DL DL $\leftarrow$ DL'

Delete instances covered by Rule from Pos and Neg

Until Pos = ϕ

Return RuleSet

PruneRuleSet(RuleSet, Pos, Neg)

For each Rule $\in$ RuleSet in reverse order

$DL \leftarrow \text{DescLen}(\text{RuleSet}, \text{Pos}, \text{Neg})$

$DL' \leftarrow \text{DescLen}(\text{RuleSet} - \text{Rule}, \text{Pos}, \text{Neg})$

If $DL' < DL$ Delete Rule from RuleSet

Return RuleSet

OptimizeRuleSet(RuleSet, Pos, Neg)

For each Rule $\in$ RuleSet

$DL0 \leftarrow \text{DescLen}(\text{RuleSet}, \text{Pos}, \text{Neg})$

$DL1 \leftarrow \text{DescLen}(\text{RuleSet} - \text{Rule} +$
 $\text{ReplaceRule}(\text{RuleSet}, \text{Pos}, \text{Neg}), \text{Pos}, \text{Neg})$

$DL2 \leftarrow \text{DescLen}(\text{RuleSet} - \text{Rule} +$
 $\text{RevisedRule}(\text{RuleSet}, \text{Rule}, \text{Pos}, \text{Neg}),$
 $\text{Pos}, \text{Neg})$

If $DL1 = \min(DL0, DL1, DL2)$

Delete Rule from RuleSet and

add $\text{ReplaceRule}(\text{RuleSet}, \text{Pos}, \text{Neg})$

Else if $DL2 = \min(DL0, DL1, DL2)$

Delete Rule from RuleSet and

add $\text{RevisedRule}(\text{RuleSet}, \text{Rule}, \text{Pos}, \text{Neg})$

Return RuleSet

Classification tree construction

Generate Tree (X)

If $\text{NodeEntropy}(X) < \theta_1$

create leaf labelled by majority class in X

Return

$i \leftarrow \text{SplitAttribute}(X)$

For each branch of x_i

Find X_i falling in branch

GenerateTree(X_i)

SplitAttribute(X)

$\text{minEnt} \leftarrow \text{MAX}$

For all attributes $i=1, 2, \dots, d$

if x_i is discrete with n values

Split X into $X_1, X_2, \dots, X_n$ by x_i

$e \leftarrow \text{SplitEntropy}(X_1, X_2, \dots, X_n)$

if $e < \text{MinEnt}$ $\text{MinEnt} \leftarrow e$; $\text{best} \leftarrow i$

Else /* x_i is numeric */

For all possible splits

Split X into X_1, X_2 on x_i

$e \leftarrow \text{SplitEntropy}(X_1, X_2)$

if $e < \text{MinEnt}$ $\text{MinEnt} \leftarrow e$; $\text{best} \leftarrow i$

Return best

RIPPER Algorithm important points

- 1) once a rule is grown and pruned, all positive and negative training examples covered by the rule are removed from the training set.
- 2) Typically, we stop if the description of the rule is not shorter than the description of instances it explains.
- 3) The description length of a rule base is the sum of the description lengths of all the rules in the rule base, plus the description of instances not covered by the rule base.
- 4) Ripper stops adding rules when the description length of the rule base is more than 64 bits larger than the best description length so far.
- 5) once the rule base is learned, we pass over the rules in reverse order to see if they can be removed without increasing the description length.
- 6) Rules in the base are also optimized after they learned. Ripper considers two alternatives to a rule: one called replacement rule, starts from an empty rule, is grown, and is then pruned. The second, called the revision rule, starts with the rule as it is, is grown and is then pruned. These two are compared with original rule, and the shortest of three is added to the rule base. This optimization of the rule base can be done K-times.

1) Use Principal Component Analysis for dimensionality reduction.

	Ex1	Ex2	Ex3	Ex4
x_1	4	8	13	7
x_2	11	4	5	14

Solution: Step 1: Find the mean

$$\bar{x}_1 = \frac{4+8+13+7}{4} = 8$$

$$\bar{x}_2 = \frac{11+4+5+14}{4} = 8.5$$

Step 2: Covariance matrix calculation

$$S = \begin{bmatrix} \text{Cov}(x_1, x_1) & \text{Cov}(x_1, x_2) \\ \text{Cov}(x_2, x_1) & \text{Cov}(x_2, x_2) \end{bmatrix}$$

$$\text{Cov}(x_1, x_1) = \frac{1}{N-1} \sum_{k=1}^N (x_{1k} - \bar{x}_1)^2$$

$$= \frac{1}{3} [(4-8)^2 + (8-8)^2 + (13-8)^2 + (7-8)^2]$$

$$= \frac{1}{3} [16 + 0 + 25 + 1] = 14$$

$$\text{Cov}(x_2, x_2) = \frac{1}{N-1} \sum_{k=1}^N (x_{2k} - \bar{x}_2)^2$$

$$= \frac{1}{3} [(11-8.5)^2 + (4-8.5)^2 + (5-8.5)^2 + (14-8.5)^2]$$

$$= \frac{1}{3} [6.25 + 20.25 + 12.25 + 30.25]$$

$$= 23$$

$$\text{Cov}(x_1, x_2) = \frac{1}{N-1} \sum_{k=1}^N [(x_{1k} - \bar{x}_1)(x_{2k} - \bar{x}_2)]$$

(2)

$$= \frac{1}{3} \left[(4-8)(11-8.5) + (8-8)(4-8.5) + (13-8)(5-8.5) + (7-8)(14-8.5) \right]$$

x_1	4	8	13	7
x_2	11	4	5	14
$\bar{x}_1$	= 8			
$\bar{x}_2$	= 8.5			

$$= \frac{1}{3} [-10 - 17.5 - 5.5] = -11$$

$$S = \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}$$

Step 3: Eigen values
 $|S - \lambda I| = 0$

$$\begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 14-\lambda & -11 \\ -11 & 23-\lambda \end{bmatrix} = 0$$

$$(14-\lambda)(23-\lambda) - 121 = 0$$

$$\lambda^2 - 37\lambda + 201 = 0$$

$$\lambda = 30.38$$

$$\lambda = 6.615$$

Step 4

Eigen vectors

$$\begin{bmatrix} 14-\lambda & -11 \\ -11 & 23 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$(14-\lambda)u_1 - 11u_2 = 0 \rightarrow (1)$$

$$-11u_1 + (23-\lambda)u_2 = 0 \rightarrow (2)$$

$$(14-\lambda)u_1 = 11u_2$$

$$\frac{u_1}{11} = \frac{u_2}{14-\lambda} = t$$

$$u_1 = 11t \quad u_2 = (14-\lambda)t$$

When $t = 1$

$$u_1 = 11 \quad u_2 = 14 - 30.38 = -16.38$$

$$\lambda = 30.38$$

$$\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 11 \\ -16.68 \end{bmatrix}$$

(3)

Step 5:

unit eigen vectors

$$e_1 = \frac{u_1}{\|u\|} = 0.55$$

$$e_2 = \frac{u_2}{\|u\|} = -0.834$$

$$= \frac{(11, -16.68)}{19.9805}$$

$$e = \begin{bmatrix} 0.55 \\ -0.834 \end{bmatrix}$$

$$\begin{array}{c|ccc} x_1 & 4 & 8 & 13 & 7 \\ \hline x_2 & 11 & 4 & 5 & 14 \end{array}$$

$$\begin{array}{c|c} & x_{11} \\ \hline & 4 \\ & 8 \end{array}$$

Step 6:

First Principal Component

For $k=1$

$$e^T \begin{bmatrix} (x_{1k} - \bar{x}_1) \\ (x_{2k} - \bar{x}_2) \end{bmatrix}$$

$$\bar{x}_1 = 8 \quad \bar{x}_2 = 8.5$$

$$\begin{bmatrix} 0.55 & -0.834 \end{bmatrix} \begin{bmatrix} (4-8) \\ (11-8.5) \end{bmatrix} = -4.285$$

$$\begin{array}{c|c} x_1 & 8 \\ \hline x_2 & 4 \end{array}$$

For $k=2$

$$e^T \begin{bmatrix} (x_{1k} - \bar{x}_1) \\ (x_{2k} - \bar{x}_2) \end{bmatrix} \Rightarrow \begin{bmatrix} 0.55 & -0.834 \end{bmatrix} \begin{bmatrix} (8-8) \\ (4-8.5) \end{bmatrix}$$

$$\bar{x}_1 = 8 \quad \bar{x}_2 = 8.5$$

$$= \begin{bmatrix} 0.55 & -0.834 \end{bmatrix} \begin{bmatrix} 0 \\ -4.5 \end{bmatrix}$$

$$= 3.753$$

For $k=3$

$$e^T \begin{bmatrix} (x_{1k} - \bar{x}_1) \\ (x_{2k} - \bar{x}_2) \end{bmatrix} \Rightarrow$$

$$\begin{array}{c|ccc} x_1 & & & 13 \\ \hline x_2 & & & 5 \end{array}$$

$$\begin{bmatrix} 0.55 & -0.834 \end{bmatrix} \begin{bmatrix} (13-8) \\ (5-8.5) \end{bmatrix} = 5.669$$

|||y

For

$k=4$

$$e^T \begin{bmatrix} (x_{1k} - \bar{x}_1) \\ (x_{2k} - \bar{x}_2) \end{bmatrix} \Rightarrow$$

$$\begin{array}{c|ccc} & & & 7 \\ \hline & & & 14 \end{array}$$

$$\begin{bmatrix} 0.55 & -0.834 \end{bmatrix} \begin{bmatrix} (7-8) \\ (14-8.5) \end{bmatrix} = -5.137$$

(4)

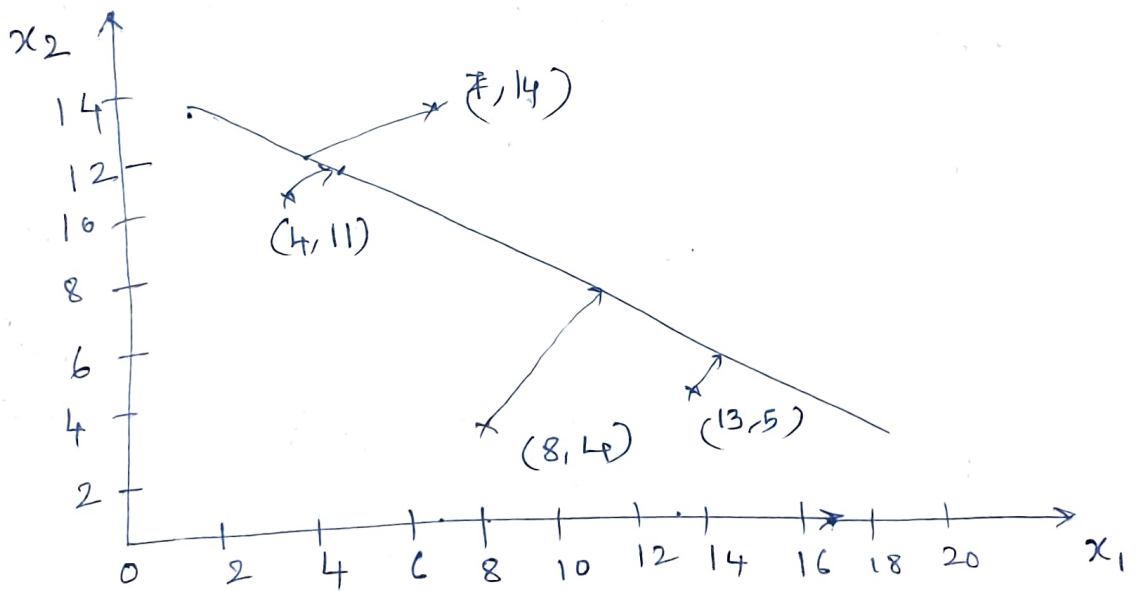
	K=1	K=2	K=3	K=4
	Ex_1	Ex_2	Ex_3	Ex_4
x_1	4	8	13	7
x_2	11	4	5	14

-4.285

3.753

5.669

-5.137



(5)

1) Find the SVD of $A = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix}$

$$A = U \Sigma V^T$$

$2 \times 2 \quad 2 \times 3 \quad 3 \times 2$

Solution:

Step 1:

$$A A^T = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix} \cdot \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 11 & 1 \\ 1 & 11 \end{bmatrix}$$

Step 2:

Eigen values and Eigen vectors of $A A^T$

$$\begin{bmatrix} 11 & 1 \\ 1 & 11 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 0$$

$$\begin{bmatrix} 11-\lambda & 1 \\ 1 & 11-\lambda \end{bmatrix} = 0$$

$$(11-\lambda)(11-\lambda) - 1 = 0$$

$$\lambda^2 - 22\lambda + 120 = 0$$

$$\lambda = 12 \quad \lambda = 10$$

$$\begin{bmatrix} 11-\lambda & 1 \\ 1 & 11-\lambda \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\lambda = 12 \Rightarrow \begin{bmatrix} 11-12 & 1 \\ 1 & 12-1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} -u_1 + u_2 &= 0 \Rightarrow u_1 = 1 \\ u_1 - u_2 &= 0 \quad u_2 = 1 \end{aligned}$$

$$\lambda = 10$$

$$\Rightarrow \begin{bmatrix} 11-10 & 1 \\ 1 & 11-10 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = 0$$

$$u_1 = 1 \quad u_2 = -1$$

Normalized $\frac{u_1}{\|u_1\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$$\frac{u_2}{\|u_2\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

Step 4:

$$A^T A \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 10 & 0 & 2 \\ 0 & 10 & 4 \\ 2 & 4 & 2 \end{bmatrix}$$

Step 5: Eigen values and Eigen vectors of $A^T A$

$$|A - \lambda I| = 0$$

$$\begin{bmatrix} 10 & 0 & 2 \\ 0 & 10 & 4 \\ 2 & 4 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{vmatrix} 10-\lambda & 0 & 2 \\ 0 & 10-\lambda & 4 \\ 2 & 4 & 2-\lambda \end{vmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$(10-\lambda)[(10-\lambda)(2-\lambda)-16] + 2[2(\lambda-10)] = 0$$

$$(10-\lambda)[20 - 10\lambda - 2\lambda + \lambda^2 - 16] + 2[2\lambda - 20] = 0$$

$$\lambda_1 = 12 \quad \lambda_2 = 10 \quad \lambda_3 = 0$$

$$\lambda_1 = 12$$

$$\begin{bmatrix} 10-\lambda & 0 & 2 \\ 0 & 10-\lambda & 4 \\ 2 & 4 & 2-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

~~continued~~

$$\begin{bmatrix} -2 & 0 & 2 \\ 0 & -2 & 4 \\ 2 & 4 & -10 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} -2x_1 + 0x_2 + 2x_3 &= 0 & x_1 &= x_3 \\ 0x_1 - 2x_2 + 4x_3 &= 0 & 2x_2 &= 4x_3 \\ 2x_1 + 4x_2 - 10x_3 &= 0 & 2x_2 &= 2x_3 \end{aligned}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

For $\lambda = 10$

$$\begin{bmatrix} 10-\lambda & 0 & 2 \\ 0 & 10-\lambda & 4 \\ 2 & 4 & 2-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -2 & 0 & 2 \\ 0 & -2 & 4 \\ 2 & 4 & -8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

My For $\lambda = 0$ $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ +2 \\ -5 \end{bmatrix}$

$$V = \begin{bmatrix} 1 & -2 & 1 \\ 2 & 1 & 2 \\ 1 & 0 & -5 \end{bmatrix}$$

Normalized $V = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{-2}{\sqrt{5}} & \frac{1}{\sqrt{30}} \\ \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{5}} & \frac{+2}{\sqrt{30}} \\ \frac{1}{\sqrt{6}} & 0 & \frac{-5}{\sqrt{30}} \end{bmatrix}$

Normalized $V^T = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{-2}{\sqrt{5}} & \frac{1}{\sqrt{5}} & 0 \\ \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{30}} & \frac{-5}{\sqrt{30}} \end{bmatrix}$

~~Sum~~ $\begin{bmatrix} \text{scribbles} \\ \text{scribbles} \\ \text{scribbles} \end{bmatrix}$

$\Sigma = \begin{bmatrix} \sqrt{12} & 0 & 0 \\ 0 & \sqrt{10} & 0 \end{bmatrix}$

$A = U \Sigma V^T$
 $2 \times 3 = 2 \times 2 \quad 2 \times 3 \quad 3 \times 3$

9

① Compute the Linear Discriminant (Analysis) projection for the 2D dataset

$$X_1 = (x_1, x_2) = \{(4, 1), (2, 4), (2, 3), (3, 6), (4, 4)\}$$

$$X_2 = (x_1, x_2) = \{(9, 10), (6, 8), (9, 5), (8, 7), (10, 8)\}$$

Solution: Step 1: Compute the mean vectors

$$\mu_1 = \frac{1}{5} \sum_{i=1}^5 x_{1i} = \frac{1}{5} [(4+2+2+3+4), (1+4+3+6+4)]$$

$$= [3, 3.6]$$

$$\mu_2 = \frac{1}{5} \sum_{i=1}^5 x_{2i} = \frac{1}{5} [(9+6+9+8+10), (10+8+5+7+8)]$$

$$= [8.4, 7.6]$$

Step 2: Compute scatter Matrices, Within class scatter matrix $S_W = S_1 + S_2$

For each class compute

$$S_i = \sum_{x \in X_i} (x - \mu_i)(x - \mu_i)^T$$

$$\text{For } X_1: S_1 = \sum_{i=1}^5 (x_{1i} - \mu_1)(x_{1i} - \mu_1)^T$$

$$(x_i - \mu_1) = \{(4-3), (1-3.6), (2-3), (4-3), (2-3)\}$$

$$(x - \mu_1) = \{(4-3, 1-3.6), (2-3, 4-3.6),$$

$$(2-3, 3-3.6), (3-3, 6-3.6), (4-3, 4-3.6)\}$$

$$(x - \mu_1) = \{(1, -2.6), (-1, 0.4), (-1, -0.6)$$

$$(0, 2.4), (1, 0.4)\}$$

$$(x - \mu_1) = \begin{bmatrix} 1 & -1 & -1 & 0 & 1 \\ -2.6 & 0.4 & -0.6 & 2.4 & 0.4 \end{bmatrix}_{2 \times 5}$$

$$(x - \mu_1)^T = \begin{bmatrix} 1 & -2.6 \\ -1 & 0.4 \\ -1 & -0.6 \\ 0 & 2.4 \\ 1 & 0.4 \end{bmatrix}_{5 \times 2}$$

$$(x - \mu_1)(x - \mu_1)^T = \begin{bmatrix} 4 & -2 \\ -2 & 13.2 \end{bmatrix}$$

(10)

Similarly

$$S_2 = \begin{bmatrix} 0.6 & -2.4 & 0.6 & -0.4 & 1.6 \\ 2.4 & 0.4 & -2.6 & -0.6 & 0.4 \end{bmatrix} \begin{bmatrix} 0.6 & 2.4 \\ -2.4 & 0.4 \\ 0.6 & -2.6 \\ -0.4 & -0.6 \\ 1.6 & 0.4 \end{bmatrix}$$

$$S_2 = \begin{bmatrix} 9.2 & -0.2 \\ -0.2 & 13.2 \end{bmatrix}$$

$$S_W = S_1 + S_2 = \begin{bmatrix} 13.2 & -2.2 \\ -2.2 & 26.4 \end{bmatrix}$$

Step 3: Compute S_W^{-1}

$$S_W^{-1} = \frac{\text{adj}(S_W)}{|S_W|} = \frac{\begin{bmatrix} 26.4 & 2.2 \\ 2.2 & 13.2 \end{bmatrix}}{343.64}$$

Step 4: $\vec{W} = S_W^{-1} (\mu_1 - \mu_2)$

$$\vec{W} = \begin{bmatrix} 0.0768 & 0.0064 \\ 0.0064 & 0.0384 \end{bmatrix} \begin{bmatrix} -5.4 \\ -4 \end{bmatrix} = \begin{bmatrix} -0.440 \\ -0.188 \end{bmatrix}$$

$$\text{Normalized } \vec{W} = W_{\text{norm}} = \begin{bmatrix} -0.44/0.479 \\ -0.188/0.479 \end{bmatrix}$$

$$= \begin{bmatrix} -0.918 \\ -0.392 \end{bmatrix}$$

Project Values

(11)

1) Scalar Projection $y = x^T W = x_1 w_1 + x_2 w_2$

2) Projection Projected 2D point on the line

$$x_{\text{proj}} = y W$$

For x_1 :

$$x_1 = \{ (4, 1), (2, 4), (2, 3), (3, 6), (4, 4) \}$$

a) (4, 1)
Scalar

$$y = 4(-0.918) + 1(-0.392) = -4.064$$

$$x_{\text{proj}} = -4.064(-0.918, -0.392) = (3.73, 1.59)$$

b) (2, 4)

$$y_{\text{scalar}} = 2(-0.918) + 4(-0.392) = -3.404$$

$$x_{\text{proj}} = -3.404(-0.918, -0.392) = (3.12, 1.33)$$

c) (2, 3)

$$y_{\text{scalar}} = 2(-0.918) + 3(-0.392) = -3.012$$

$$x_{\text{proj}} = -3.012(-0.918, -0.392) = (2.77, 1.18)$$

d) (3, 6)

$$y_{\text{scalar}} = 3(-0.918) + 6(-0.392) = -5.106$$

$$x_{\text{proj}} = -5.106(-0.918, -0.392) = (4.69, 2)$$

e) (4, 4)

$$y_{\text{scalar}} = 4(-0.918) + 4(-0.392) = -5.24$$

$$x_{\text{proj}} = -5.24(-0.918, -0.392) = (4.81, 2.05)$$

For x_2 :

$$x_2 = \{ (9, 10), (6, 8), (9, 5), (8, 7), (10, 8) \}$$

a) (9, 10)

$$y_{\text{scalar}} = 9(-0.918) + 10(-0.392) = -12.182$$

$$x_{\text{proj}} = -12.182(-0.918, -0.392) = (11.18, 4.77)$$

b) (6, 8)

$$y_{\text{scalar}} = 6(-0.918) + 8(-0.392) = -8.644$$

$$x_{\text{proj}} = -8.644(-0.918, -0.392) = (7.93, 3.39)$$

$$c) (9, 5) \quad y_{\text{scalar}} = -10.222$$

$$x_{\text{proj}} = (9.38, 4)$$

$$d) (8, 7) \quad y_{\text{scalar}} = -10.088$$

$$x_{\text{proj}} = (9.26, 3.95)$$

$$e) (10, 8) \quad y_{\text{scalar}} = -12.316$$

$$x_{\text{proj}} = (11.31, 4.82)$$

$$\vec{W} = (-0.918, -0.392)$$

$$\text{slope} = \frac{W_2}{W_1} = \frac{-0.392}{-0.918} = 0.43$$

Point to draw from midpoint of class members

$$\mu_1 = (3, 3.6) \quad \mu_2 = (8.4, 7.6)$$

$$\text{mid point} = (5.7, 5.6)$$

Generate 2 points using $\vec{W}$

$$(x, y) = \text{midpoint} + t \cdot \vec{W}$$

$$\text{If } t = 2$$

$$(x, y) = (5.7, 5.6) + 2(-0.918, -0.392)$$

$$= (3.86, 4.82)$$

$$\text{If } t = -2$$

$$(x, y) = (5.7, 5.6) - 2(-0.918, -0.392)$$

$$= (7.54, 6.38)$$

for arrow start at the mid point and add $\vec{W}$

$$(5.7 - 0.918, 5.6 - 0.392) = (4.78, 5.2)$$

13

